

Reviewer Pack — Schur-Weyl Duality and Permanent vs Determinant Circuit Lowe...

Entry e78d7cc91f22 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Schur-Weyl Duality and Permanent vs Determinant Circuit Lower Bounds

Entry ID: e78d7cc91f22 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-21 19:47:20 UTC

1. Conjecture

Field A (mathematical branch): Schur-Weyl duality **Field B** (complexity object): Permanent vs determinant circuit complexity

Statement:

{‘sentence_1’: ‘The Schur-Weyl coefficients of a given homogeneous polynomial f in two variables with degree n are computed.’, ‘sentence_2’: ‘The permanent and determinant polynomials of f are defined as the respective traces of matrices formed by these coefficients.’, ‘sentence_3’: ‘For all homogeneous polynomials f , if the Schur-Weyl coefficients are non-zero for all $m < n^{1.5}$, then the circuit complexity of computing the permanent is at least that of the determinant raised to the power $O(1)$.’}

Rationale (proposer’s reasoning):

{‘sentence_1’: ‘Schur-Weyl duality provides a bridge between representation theory and the symmetric functions that could reveal deeper connections between these mathematical objects.’, ‘sentence_2’: ‘The permanent vs determinant problem has been a central question in complexity theory, with potential implications for proving $P \neq NP$.’, ‘sentence_3’: ‘This conjecture suggests a new way to use representation-theoretic tools to gain insights into the circuit complexity of fundamental problems.’}

Taxonomy category: AVG_TO_WORST_CASE (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 0fa2d2b97217ddff

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all $n \leq 40$, at least 80% of the generated polynomials f have non-zero Schur-Weyl coefficients for $m < n^{1.5}$ and the circuit complexity of computing the permanent is within $O(1)$ times that of the determinant.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'Schur-Weyl duality' AND 'permanent vs determinant circuit complexity' AND 'circuit lower bounds' - 'homogeneous polynomial' AND 'Schur-Weyl coefficients' AND 'permanent determinant trace' - 'matrix formation' AND 'Schur-Weyl' AND 'determinant raised to power 0(1)'

Top relevant hits considered: - [<http://arxiv.org/abs/1302.4494v2>] Standard multipartitions and a combinatorial affine Schur-Weyl duality - [<http://arxiv.org/abs/cs/0512008v2>] Cohomology in Grothendieck Topologies and Lower Bounds in Boolean Complexity - [<http://arxiv.org/abs/2604.23700v1>] Quantum Circuit Cutting: Complexity and Optimization - [<http://arxiv.org/abs/1311.0820v1>] Polynomial representations of GL(n) and Schur-Weyl duality - [<http://arxiv.org/abs/1407.3582v2>] Even dimensional homogeneous Finsler spaces with positive flag curvature - [<http://arxiv.org/abs/2510.06739v2>] Asymptotics of the Hankel determinant and orthogonal polynomials arising from the information theory of MIMO systems - [<http://arxiv.org/abs/1005.0979v1>] Supersymmetry in Random Matrix Theory - [<http://arxiv.org/abs/math/0610498v4>] Bounds on changes in Ritz values for a perturbed invariant subspace of a Hermitian matrix

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_polynomial(n):
        coefficients = [random.randint(-10, 10) for _ in range(n)]
        return coefficients

    def compute_schur_weyl_coefficients(poly, m):
        n = len(poly)
        if m >= n**1.5:
            return []
        # Placeholder implementation
        return [poly[i] * poly[j] for i in range(m+1) for j in range(i, m+1)]

    def compute_permanent(poly):
```

```

n = len(poly)
permanent = 0
for perm in itertools.permutations(range(n)):
    product = 1
    for i in range(n):
        product *= poly[perm[i]]
    permanent += product
return permanent

def compute_determinant(poly):
    n = len(poly)
    if n == 1:
        return poly[0]
    det = 0
    for j in range(n):
        sub_poly = [poly[i] for i in range(1, n) if i != j]
        det += (-1)**j * poly[j] * compute_determinant(sub_poly)
    return det

def circuit_complexity(poly):
    # Placeholder implementation
    return len(poly)

n_values = [5, 10, 15, 20, 30, 40]
results = []

for n in n_values:
    poly = generate_polynomial(n)
    schur_weyl_coeffs = compute_schur_weyl_coefficients(poly, int(n**1.5))

    if not all(coeff != 0 for coeff in schur_weyl_coeffs):
        return {
            "metric_name": "Schur-Weyl Coefficients",
            "metric_value": None,
            "instances_tested": len(schur_weyl_coeffs),
            "conjecture_holds": False,
            "counterexample": "Non-zero Schur-Weyl coefficients not found for  $m < n^{1.5}$ "
        }

    permanent = compute_permanent(poly)
    determinant = compute_determinant(poly)
    perm_complexity = circuit_complexity(permanent)
    det_complexity = circuit_complexity(determinant)

    results.append({
        "n": n,
        "permanent": permanent,
        "determinant": determinant,
        "perm_complexity": perm_complexity,
        "det_complexity": det_complexity
    })

total_perm_complexity = sum(result["perm_complexity"] for result in results)
total_det_complexity = sum(result["det_complexity"] for result in results)

return {

```

```

        "metric_name": "Circuit Complexity Ratio",
        "metric_value": total_perm_complexity / total_det_complexity,
        "instances_tested": len(results),
        "conjecture_holds": all(result["perm_complexity"] >= result["det_complexity"] for result in results),
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    total_perm_complexity = sum(result["perm_complexity"] for result in results)
    total_det_complexity = sum(result["det_complexity"] for result in results)
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={total_perm_complexity/total_det_complexity} std=0.0 support={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={total_perm_complexity/total_det_complexity} std=0.0 support={support_fraction}")
    else:
        for result in results:
            if not result["conjecture_holds"]:
                counterexample = f"n={result['n']}, permanent={result['permanent']}, determinant={result['determinant']}"
                print(f"RESULT: FALSIFIED counterexample={counterexample} first_failing_seed={seed}")
                break

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9eb4f3b9.py", line 103, in <module>
    trial_result = run_trial(seed)
                   ~~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9eb4f3b9.py", line 61, in run_trial
    schur_weyl_coeffs = compute_schur_weyl_coefficients(poly, int(n**1.5))
                        ~~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9eb4f3b9.py", line 30, in compute_schur
    return [poly[i] * poly[j] for i in range(m+1) for j in range(i, m+1)]
           ~~~~~~
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete the required computations to verify the conjecture. | next: Re-run the test with proper error handling and ensure that it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14966
2	preregistration	ollama_remote	glm4:latest	0	0	9502
3	novelty	ollama_remote	glm4:latest	0	0	8555
4	novelty	ollama_remote	glm4:latest	0	0	10106
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13918
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9805
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10961
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11671
9	judge	ollama_remote	glm4:latest	0	0	12882

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 102366 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/e78d7cc91f22.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/e78d7cc91f22.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/e78d7cc91f22.tar.gz` (if generated)